

Gaurav Mukherjee

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 71% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, FreeRTOS, ARM Cortex, C/C++, PCB Debugging, MQTT, I2C/SPI/UART

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 54%.
- Led a squad of 4 engineers delivering scalable microservices processing 12M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 70%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2022 - 2023)

- Designed and maintained FreeRTOS data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$40k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Embedded Linux, FreeRTOS, ARM Cortex, C/C++]
Engineered end-to-end production solution leveraging Embedded Linux and FreeRTOS with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [ARM Cortex, C/C++, PCB Debugging, MQTT]
Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

M.Tech in Data Science & Artificial Intelligence

ETH Zurich — 2018 - 2022 (Score: 9.28 / 10.0)